Freezing Strategies in Transfer Learning for Skin Lesion Classification

Deep Learning Project Report S26

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1 Introduction and Background

Transfer learning has become a key method in deep learning, allowing models trained on large datasets to be efficiently adapted to new domains where data is often limited. A central question in this process is how much of the previously learned knowledge should be preserved, frozen, or fine-tuned when moving to a different task.

This project examines different strategies for freezing layers in transfer learning applied to the classification of skin lesions, a task of clear clinical importance, where labeled data are scarce and accurate predictions can have direct diagnostic value. By systematically testing various freezing configurations, we aim to better understand the balance between preserving general visual representations and adapting to task-specific features.

We take inspiration from an analogy in medical training: when a general practitioner specializes in dermatology, they keep their foundational understanding of medicine (such as anatomy and physiology) but develop new, focused expertise. In a similar way, a pre-trained neural network retains basic visual patterns like edges and textures that remain useful, while its higher-level layers often need to be fine-tuned to capture the subtle distinctions between different kinds of skin lesions.

"When adapting a pre-trained model to a new task, how much of its previously learned knowledge should be retained?"

2 Methodology

We will use the HAM10000 dataset (Human Against Machine with 10,000 training images), accessible via Kaggle and TensorFlow Datasets. Key dataset properties:

- 10,015 dermoscopic images across 7 classes: melanoma (MEL), melanocytic nevi (NV), basal cell carcinoma (BCC), actinic keratosis (AKIEC), benign keratosis (BKL), dermatofibroma (DF), and vascular lesions (VASC)
- Significant class imbalance (NV dominates with ~67% of samples)
- Standard train/validation/test splits will be applied with stratified sampling
- Preprocessing: resize to 224×224, ImageNet normalization, augmentation (flip, rotation, color jitter)

Due to class imbalance, we will apply class-weighted loss during training and report balanced accuracy and macro-averaged AUC as primary metrics.

2.1 Base Architecture

We will use EfficientNet-B0 pre-trained on ImageNet as our primary backbone, due to its favorable accuracy-efficiency trade-off and widespread use in medical imaging benchmarks. ResNet-50 will be used for cross-architecture validation in a subset of experiments.

We compare three freezing strategies:

Strategy	Description	Expected Behaviour
S1 – From Scratch	All weights randomly initialised; full network trained on HAM10000	Weakest performance; lower bound baseline due to limited data
S2 – Full Freeze	All pre-trained weights frozen; only the classification head is trained	Fast training; may underfit if ImageNet features are insufficient
S3 – Gradual Unfreeze	Start with frozen backbone; progressively unfreeze layer blocks every few epochs	Best generalisation; balances retained knowledge with task-specific adaptation

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